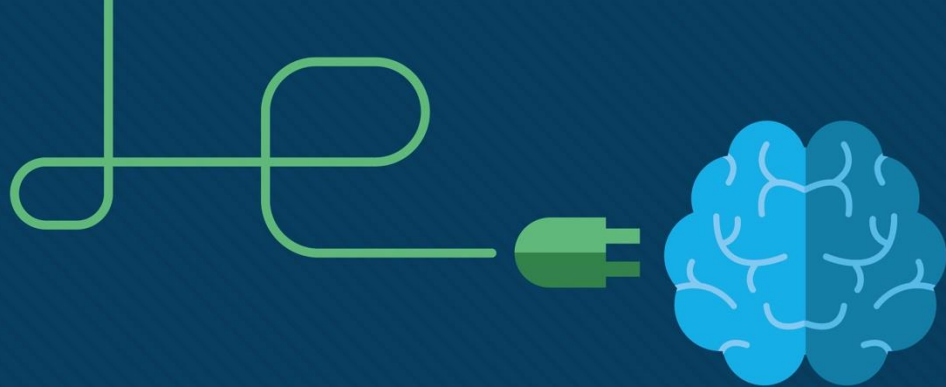




Module 11: IPv4 Addressing

Introduction to Networks v7.0
(ITN)

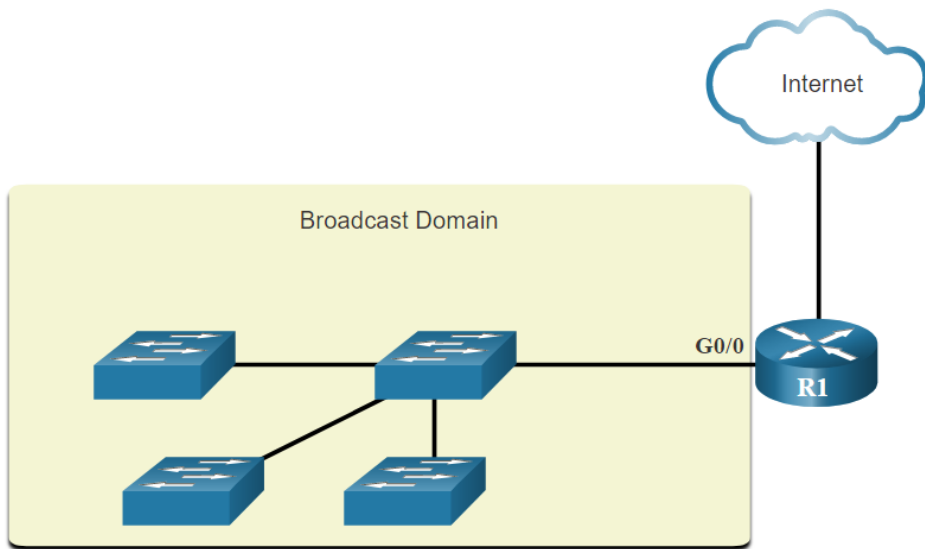


Network Segmentation

Network Segmentation

Broadcast Domains and Segmentation

- Many protocols use broadcasts or multicasts (e.g., ARP use broadcasts to locate other devices, hosts send DHCP discover broadcasts to locate a DHCP server.)
- Switches propagate broadcasts out all interfaces except the interface on which it was received.

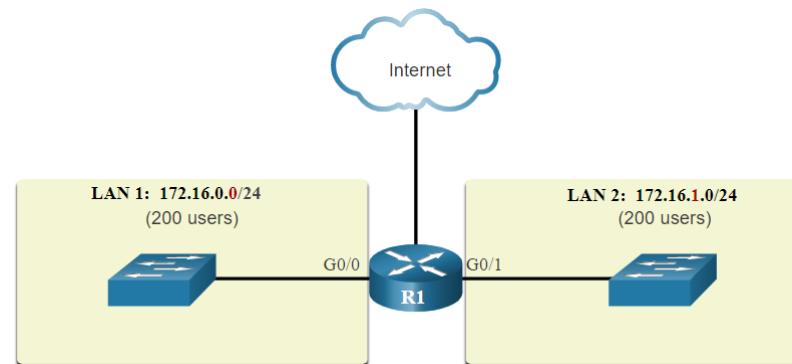
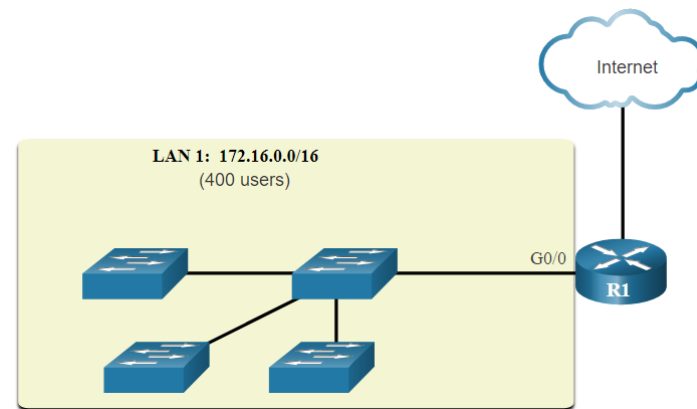


- The only device that stops broadcasts is a router.
- Routers do not propagate broadcasts.
- Each router interface connects to a broadcast domain and broadcasts are only propagated within that specific broadcast domain.

Network Segmentation

Problems with Large Broadcast Domains

- A problem with a large broadcast domain is that these hosts can generate excessive broadcasts and negatively affect the network.
- The solution is to reduce the size of the network to create smaller broadcast domains in a process called subnetting.
- Dividing the network address 172.16.0.0 /16 into two subnets of 200 users each: 172.16.0.0 /24 and 172.16.1.0 /24.
- Broadcasts are only propagated within the smaller broadcast domains.

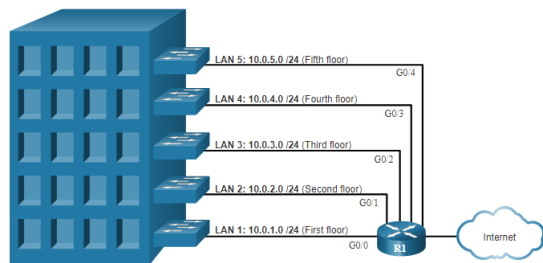


Network Segmentation

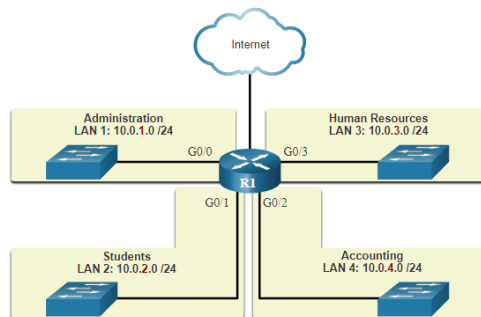
Reasons for Segmenting Networks

- Subnetting reduces overall network traffic and improves network performance.
- It can be used to implement security policies between subnets.
- Subnetting reduces the number of devices affected by abnormal broadcast traffic.
- Subnets are used for a variety of reasons including by:

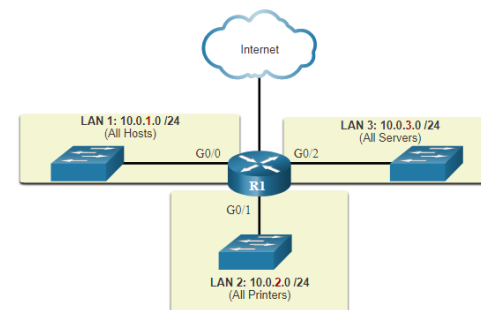
Location



Group or Function



Device Type

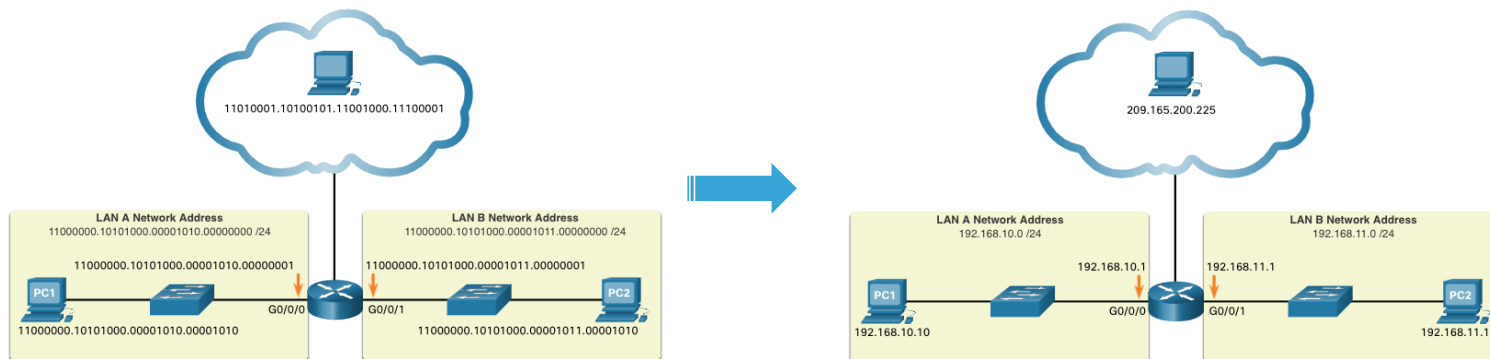


Binary Number System

Binary Number System

Binary and IPv4 Addresses

- Binary numbering system consists of 1s and 0s, called bits
- Decimal numbering system consists of digits 0 through 9
- Hosts, servers, and network equipment using binary addressing to identify each other.
- Each address is made up of a string of 32 bits, divided into four sections called octets.
- Each octet contains 8 bits (or 1 byte) separated by a dot.
- For ease of use by people, this dotted notation is converted to dotted decimal.



Binary Number System

Binary Positional Notation

- Positional notation means that a digit represents different values depending on the “position” the digit occupies in the sequence of numbers.
- The decimal positional notation system operates as shown in the tables below.

Radix	10	10	10	10
Position in Number	3	2	1	0
Calculate	(10^3)	(10^2)	(10^1)	(10^0)
Position Value	1000	100	10	1



	Thousands	Hundreds	Tens	Ones
Positional Value	1000	100	10	1
Decimal Number (1234)	1	2	3	4
Calculate	1×1000	2×100	3×10	4×1
Add them up...	1000	+ 200	+ 30	+ 4
Result	1,234			

Binary Number System

Binary Positional Notation (Cont.)

The binary positional notation system operates as shown in the tables below.

Radix	2	2	2	2	2	2	2	2
Position in Number	7	6	5	4	3	2	1	0
Calculate	(2^7)	(2^6)	(2^5)	(2^4)	(2^3)	(2^2)	(2^1)	(2^0)
Position Value	128	64	32	16	8	4	2	1



Positional Value	128	64	32	16	8	4	2	1
Binary Number (11000000)	1	1	0	0	0	0	0	0
Calculate	1x128	1x64	0x32	0x16	0x8	0x4	0x2	0x1
Add Them Up...	128	+ 64	+ 0	+ 0	+ 0	+ 0	+ 0	+ 0
Result	192							

Binary Number System

Convert Binary to Decimal

Convert 11000000.10101000.00001011.00001010 to decimal.

Positional Value	128	64	32	16	8	4	2	1
Binary Number (11000000)	1	1	0	0	0	0	0	0
Calculate	1x128	1x64	0x32	0x16	0x8	0x4	0x2	0x1
Add Them Up...	128	+ 64	+ 0	+ 0	+ 0	+ 0	+ 0	+ 0
Binary Number (10101000)	1	0	1	0	1	0	0	0
Calculate	1x128	0x64	1x32	0x16	1x8	0x4	0x2	0x1
Add Them Up...	128	+ 0	+ 32	+ 0	+ 8	+ 0	+ 0	+ 0
Binary Number (00001011)	0	0	0	0	1	0	1	1
Calculate	0x128	0x64	0x32	0x16	1x8	0x4	1x2	1x1
Add Them Up...	0	+ 0	+ 0	+ 0	+ 8	+ 0	+ 2	+ 1
Binary Number (00001010)	0	0	0	0	1	0	1	0
Calculate	0x128	0x64	0x32	0x16	1x8	0x4	1x2	0x1
Add Them Up...	0	+ 0	+ 0	+ 0	+ 8	+ 0	+ 2	+ 0

➡ 192

➡ 168

➡ 11

➡ 10

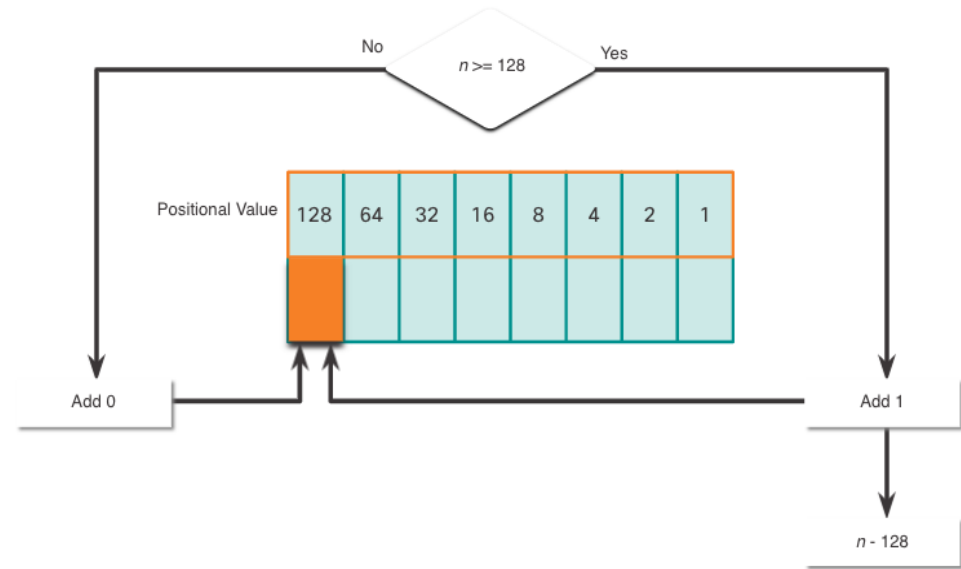
192.168.11.10

Binary Number System

Decimal to Binary Conversion

The binary positional value table is useful in converting a dotted decimal IPv4 address to binary.

- Start in the 128 position (the most significant bit). Is the decimal number of the octet (n) equal to or greater than 128?
- If no, record a binary 0 in the 128 positional value and move to the 64 positional value.
- If yes, record a binary 1 in the 128 positional value, subtract 128 from the decimal number, and move to the 64 positional value.
- Repeat these steps through the 1 positional value.



Decimal to Binary Conversion Example

- Convert decimal 168 to binary

Is $168 > 128$?

- Yes, enter 1 in 128 position and subtract 128 ($168-128=40$)

Is $40 > 64$?

- No, enter 0 in 64 position and move on

Is $40 > 32$?

- Yes, enter 1 in 32 position and subtract 32 ($40-32=8$)

Is $8 > 16$?

- No, enter 0 in 16 position and move on

Is $8 > 8$?

- Equal. Enter 1 in 8 position and subtract 8 ($8-8=0$)

No values left. Enter 0 in remaining binary positions

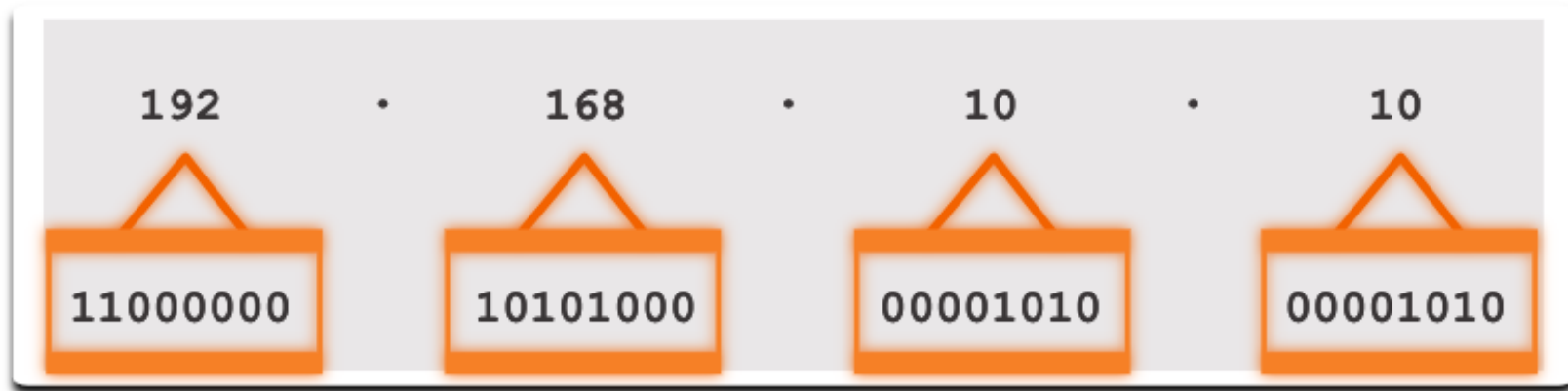
128	64	32	16	8	4	2	1
1	0	1	0	1	0	0	0

Decimal 168 is written as 10101000 in binary

Binary Number System

IPv4 Addresses

- Routers and computers only understand binary, while humans work in decimal. It is important for you to gain a thorough understanding of these two numbering systems and how they are used in networking.



Refer IP Addressing eBook

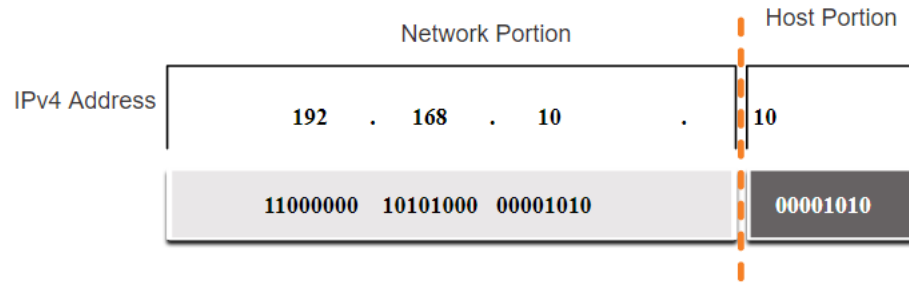
- Binary to decimal conversion at page 2
- <https://www.youtube.com/watch?v=SDG3exslzio>
- Decimal to binary conversion at page 3
- https://www.youtube.com/watch?v=dR2nNOLV_1s

IPv4 Address Structure

IPv4 Address Structure

Network and Host Portions

- An IPv4 address is a 32-bit hierarchical address that is made up of a network portion and a host portion.
- When determining the network portion versus the host portion, you must look at the 32-bit stream.
- A subnet mask is used to determine the network and host portions.



IPv4 Address Structure

The Subnet Mask

- To identify the network and host portions of an IPv4 address, the subnet mask is compared to the IPv4 address bit for bit, from left to right.
- The actual process used to identify the network and host portions is called ANDing.

	Network Portion				Host Portion
IPv4 Address	192	.	168	.	10
	11000000	10101000	00001010		00001010
Subnet Mask	255	.	255	.	0
	11111111	11111111	11111111		00000000

IPv4 Address Structure

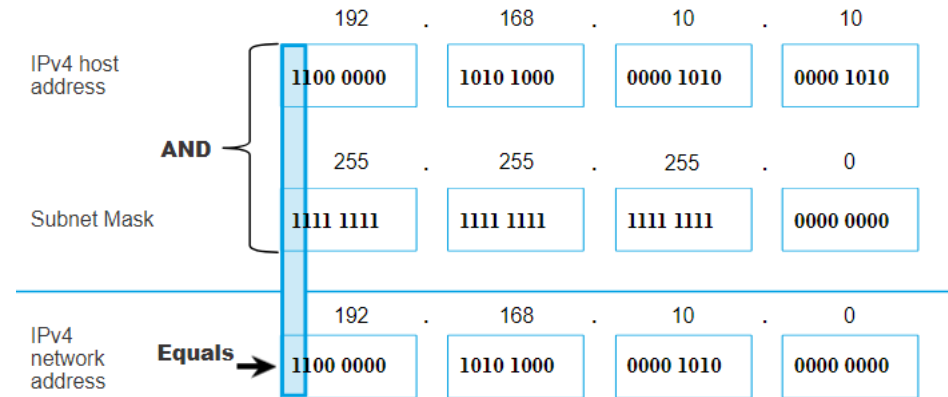
The Prefix Length

- A prefix length is a less cumbersome method used to identify a subnet mask address.
- The prefix length is the number of bits set to 1 in the subnet mask.
- It is written in “slash notation” therefore, count the number of bits in the subnet mask and prepend it with a slash.

Subnet Mask	32-bit Address	Prefix Length
255.0.0.0	11111111.00000000.00000000.00000000	/8
255.255.0.0	11111111.11111111.00000000.00000000	/16
255.255.255.0	11111111.11111111.11111111.00000000	/24
255.255.255.128	11111111.11111111.11111111.10000000	/25
255.255.255.192	11111111.11111111.11111111.11000000	/26
255.255.255.224	11111111.11111111.11111111.11100000	/27
255.255.255.240	11111111.11111111.11111111.11110000	/28
255.255.255.248	11111111.11111111.11111111.11111000	/29
255.255.255.252	11111111.11111111.11111111.11111100	/30

Determining the Network: Logical AND

- A logical AND Boolean operation is used in determining the network address.
- Logical AND is the comparison of two bits where only a 1 AND 1 produces a 1 and any other combination results in a 0.
- $1 \text{ AND } 1 = 1$, $0 \text{ AND } 1 = 0$, $1 \text{ AND } 0 = 0$, $0 \text{ AND } 0 = 0$
- $1 = \text{True}$ and $0 = \text{False}$
- To identify the network address, the host IPv4 address is logically ANDed, bit by bit, with the subnet mask to identify the network address.



Video – Network, Host and Broadcast Addresses

This video will cover the following:

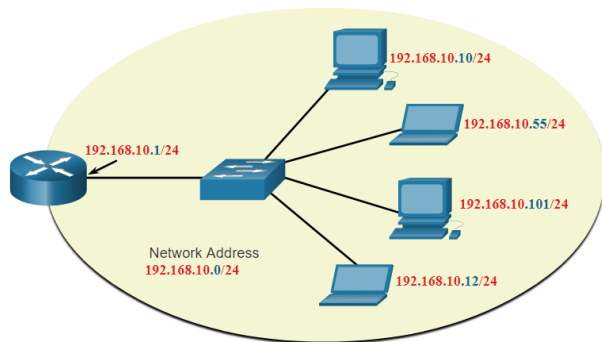
- Network address
- Broadcast Address
- First usable host
- Last usable host

11	IPv4 Addressing	^
11.0.2	What will I learn to do in this module?	
11.1	IPv4 Address Structure	^
11.1.1	Network and Host Portions	
11.1.2	The Subnet Mask	
11.1.3	The Prefix Length	
11.1.4	Determining the Network: Logical AND	
11.1.5	Video - Network, Host and Broadcast Addresses	

IPv4 Address Structure

Network, Host, and Broadcast Addresses

- Within each network are three types of IP addresses:
 - Network address
 - Host addresses
 - Broadcast address



	Network Portion			Host Portion	Host Bits
Subnet mask 255.255.255.0 or /24	255 11111111	255 11111111	255 11111111	0 00000000	
Network address 192.168.10.0 or /24	192 11000000	168 10100000	10 00001010	0 00000000	All 0s
First address 192.168.10.1 or /24	192 11000000	168 10100000	10 00001010	1 00000001	All 0s and a 1
Last address 192.168.10.254 or /24	192 11000000	168 10100000	10 00001010	254 11111110	All 1s and a 0
Broadcast address 192.168.10.255 or /24	192 11000000	168 10100000	10 00001010	255 11111111	All 1s

Refer IP Addressing eBook

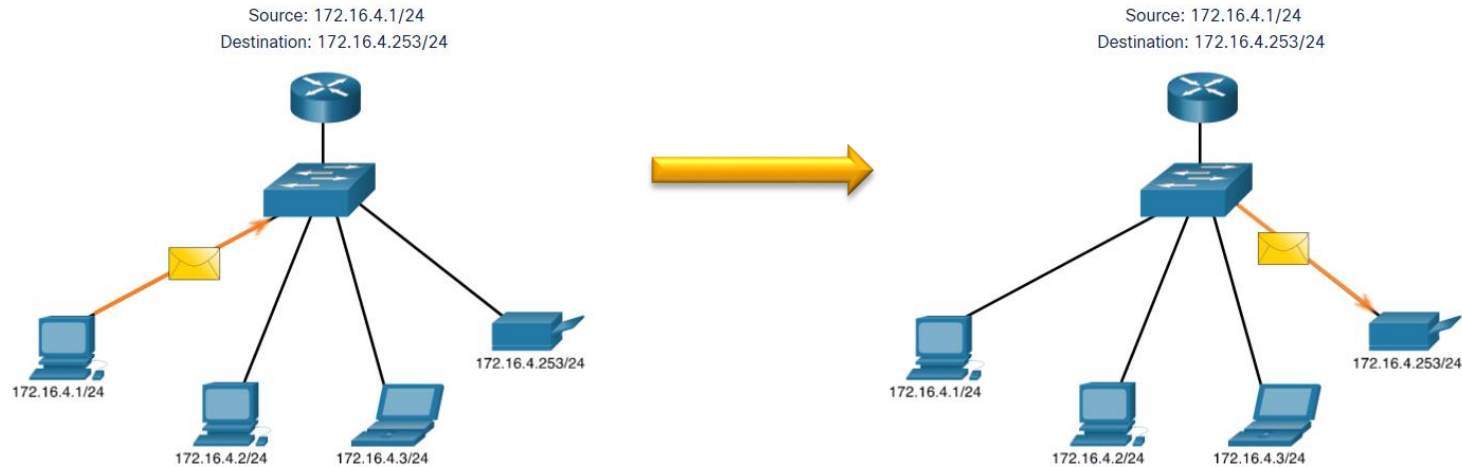
- Network and Host Identification at page 3
 - Network Addresses at page 4
 - Host Addresses at page 4
 - Default Subnet Masks at page 5
 - Custom Subnet Masks at page 6 – 8
-
- **IP address network and host portion | subnet**
 - <https://www.youtube.com/watch?v=eHV1aOnu7oM>
 - **How to calculate the total number of the usable host**
 - <https://www.youtube.com/watch?v=9uC-6lV1xT4> s?

11.2 IPv4 Unicast, Broadcast, and Multicast

IPv4 Unicast, Broadcast, and Multicast

Unicast

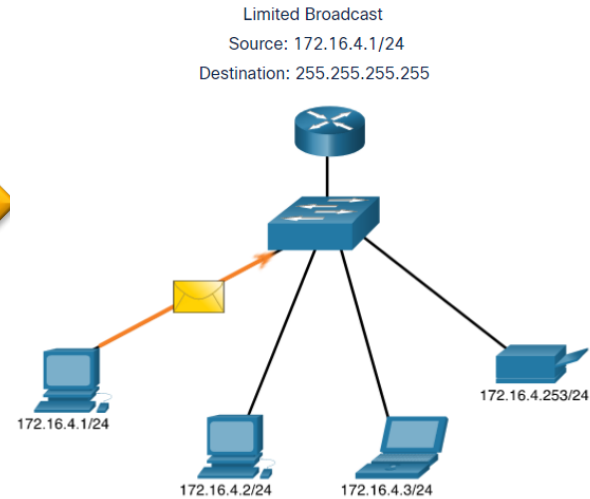
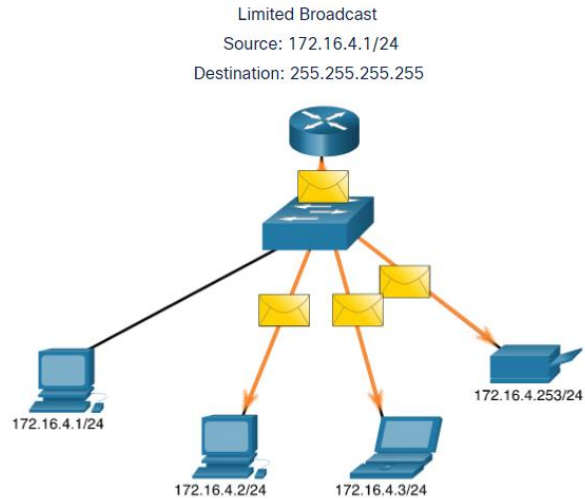
- Unicast transmission is sending a packet to one destination IP address.
- For example, the PC at 172.16.4.1 sends a unicast packet to the printer at 172.16.4.253.



IPv4 Unicast, Broadcast, and Multicast

Broadcast

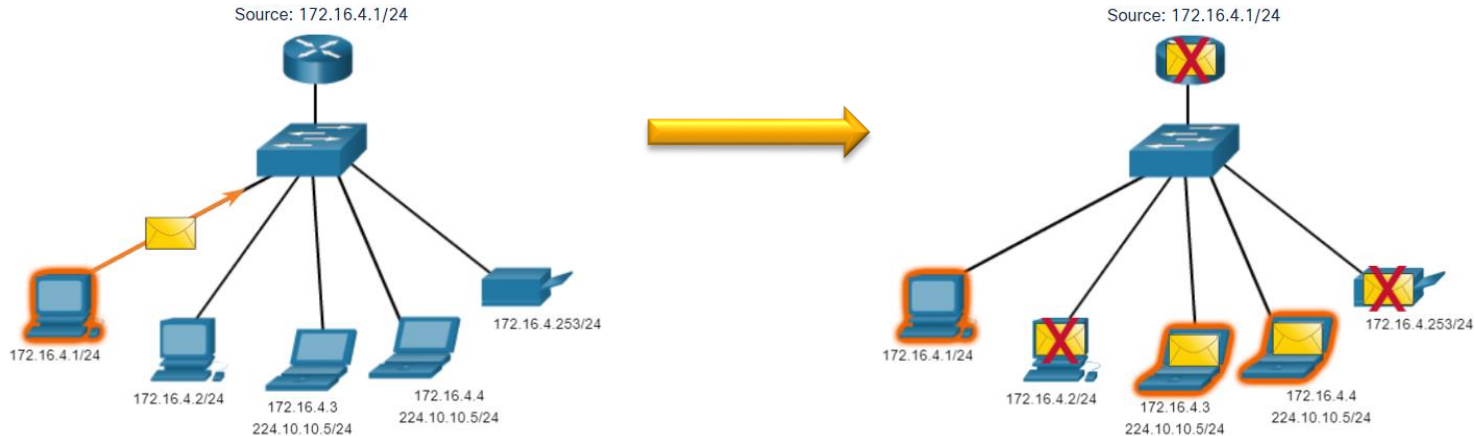
- Broadcast transmission is sending a packet to all other destination IP addresses.
- For example, the PC at 172.16.4.1 sends a broadcast packet to all IPv4 hosts.



IPv4 Unicast, Broadcast, and Multicast

Multicast

- Multicast transmission is sending a packet to a multicast address group.
- For example, the PC at 172.16.4.1 sends a multicast packet to the multicast group address 224.10.10.5.



11.3 Types of IPv4 Addresses

Public and Private IPv4 Addresses

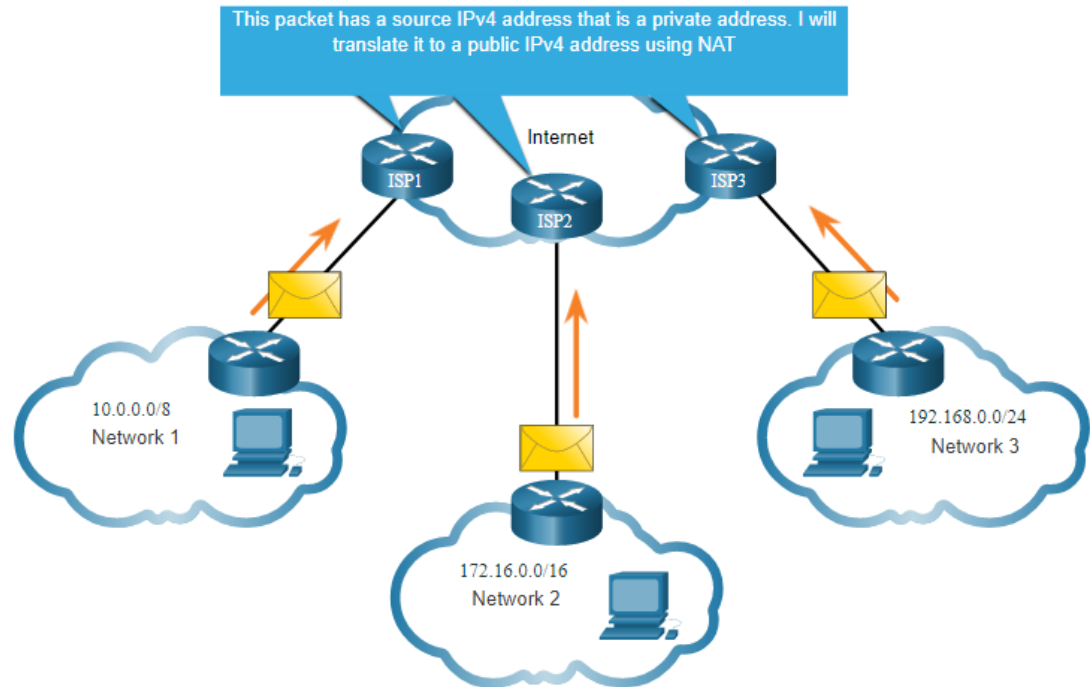
- As defined in in RFC 1918, public IPv4 addresses are globally routed between internet service provider (ISP) routers.
- Private addresses are common blocks of addresses used by most organizations to assign IPv4 addresses to internal hosts.
- Private IPv4 addresses are not unique and can be used internally within any network.
- However, private addresses are not globally routable.

Network Address and Prefix	RFC 1918 Private Address Range
10.0.0.0/8	10.0.0.0 - 10.255.255.255
172.16.0.0/12	172.16.0.0 - 172.31.255.255
192.168.0.0/16	192.168.0.0 - 192.168.255.255

Types of IPv4 Addresses

Routing to the Internet

- Network Address Translation (NAT) translates private IPv4 addresses to public IPv4 addresses.
- NAT is typically enabled on the edge router connecting to the internet.
- It translates the internal private address to a public global IP address.



Special Use IPv4 Addresses

Loopback addresses

- 127.0.0.0 /8 (127.0.0.1 to 127.255.255.254)
- Commonly identified as only 127.0.0.1
- Used on a host to test if TCP/IP is operational.

```
C:\Users\NetAcad> ping 127.0.0.1
Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
```

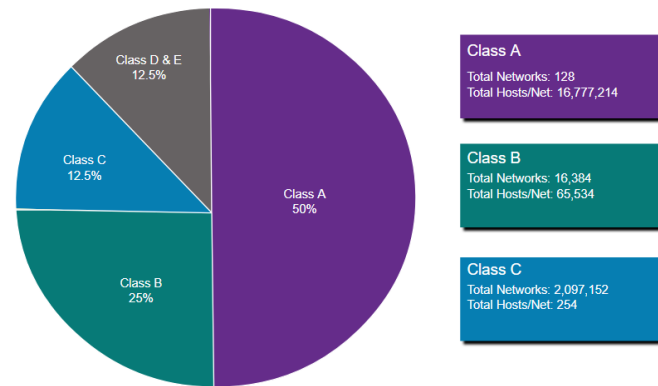
Link-Local addresses

- 169.254.0.0 /16 (169.254.0.1 to 169.254.255.254)
- Commonly known as the Automatic Private IP Addressing (APIPA) addresses or self-assigned addresses.
- Used by Windows DHCP clients to self-configure when no DHCP servers are available.

Legacy Classful Addressing

RFC 790 (1981) allocated IPv4 addresses in classes

- Class A (0.0.0.0/8 to 127.0.0.0/8)
 - Class B (128.0.0.0 /16 – 191.255.0.0 /16)
 - Class C (192.0.0.0 /24 – 223.255.255.0 /24)
 - Class D (224.0.0.0 to 239.0.0.0)
 - Class E (240.0.0.0 – 255.0.0.0)
-
- Classful addressing wasted many IPv4 addresses.

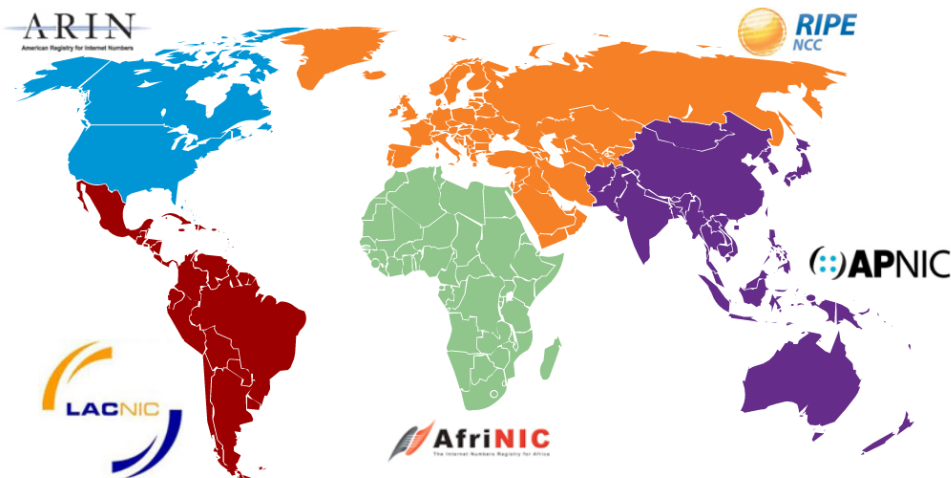


Classful address allocation was replaced with classless addressing which ignores the rules of classes (A, B, C).

Types of IPv4 Addresses

Assignment of IP Addresses

- The Internet Assigned Numbers Authority (IANA) manages and allocates blocks of IPv4 and IPv6 addresses to five Regional Internet Registries (RIRs).
- RIRs are responsible for allocating IP addresses to ISPs who provide IPv4 address blocks to smaller ISPs and organizations.



Refer IP Addressing eBook

- Address Class Identification page 11

Subnet an IPv4 Network

Subnet on an Octet Boundary

- Networks are most easily subnetted at the octet boundary of /8, /16, and /24.
- Notice that using longer prefix lengths decreases the number of hosts per subnet.

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of hosts
/8	255.0.0.0	<code>nnnnnnnnn.hhhhhhhh.hhhhhhhh.hhhhhhhh</code> <code>11111111.00000000.00000000.00000000</code>	16,777,214
/16	255.255.0.0	<code>nnnnnnnnn.nnnnnnnnn.hhhhhhhh.hhhhhhhh</code> <code>11111111.11111111.00000000.00000000</code>	65,534
/24	255.255.255.0	<code>nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.hhhhhhhh</code> <code>11111111.11111111.11111111.00000000</code>	254

Subnet an IPv4 Network

Subnet on an Octet Boundary (Cont.)

- In the first table 10.0.0.0/8 is subnetted using /16 and in the second table, a /24 mask.

Subnet Address (256 Possible Subnets)	Host Range (65,534 possible hosts per subnet)	Broadcast
10.0.0.0/16	10.0.0.1 - 10.0.255.254	10.0.255.255
10.1.0.0/16	10.1.0.1 - 10.1.255.254	10.1.255.255
10.2.0.0/16	10.2.0.1 - 10.2.255.254	10.2.255.255
10.3.0.0/16	10.3.0.1 - 10.3.255.254	10.3.255.255
10.4.0.0/16	10.4.0.1 - 10.4.255.254	10.4.255.255
10.5.0.0/16	10.5.0.1 - 10.5.255.254	10.5.255.255
10.6.0.0/16	10.6.0.1 - 10.6.255.254	10.6.255.255
10.7.0.0/16	10.7.0.1 - 10.7.255.254	10.7.255.255
...	...	...
10.255.0.0/16	10.255.0.1 - 10.255.255.254	10.255.255.255

Subnet Address (65,536 Possible Subnets)	Host Range (254 possible hosts per subnet)	Broadcast
10.0.0.0/24	10.0.0.1 - 10.0.0.254	10.0.0.255
10.0.1.0/24	10.0.1.1 - 10.0.1.254	10.0.1.255
10.0.2.0/24	10.0.2.1 - 10.0.2.254	10.0.2.255
...	...	...
10.0.255.0/24	10.0.255.1 - 10.0.255.254	10.0.255.255
10.1.0.0/24	10.1.0.1 - 10.1.0.254	10.1.0.255
10.1.1.0/24	10.1.1.1 - 10.1.1.254	10.1.1.255
10.1.2.0/24	10.1.2.1 - 10.1.2.254	10.1.2.255
...	...	...
10.100.0.0/24	10.100.0.1 - 10.100.0.254	10.100.0.255
...	...	...
10.255.255.0/24	10.255.255.1 - 10.255.255.254	10.255.255.255

Subnet an IPv4 Network

Subnet within an Octet Boundary

- Refer to the table to see six ways to subnet a /24 network.

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts
/25	255.255.255.128	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnnnnnn 11111111.11111111.11111111.10000000	2	126
/26	255.255.255.192	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnnnnnn 11111111.11111111.11111111.11000000	4	62
/27	255.255.255.224	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnnnnnn 11111111.11111111.11111111.11100000	8	30
/28	255.255.255.240	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnnnnnn 11111111.11111111.11111111.11110000	16	14
/29	255.255.255.248	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnnnnnn 11111111.11111111.11111111.11111000	32	6
/30	255.255.255.252	nnnnnnnnn.nnnnnnnnn.nnnnnnnnn.nnnnnnnnn 11111111.11111111.11111111.11111100	64	2

Video – The Subnet Mask

- This video will demonstrate the process of subnetting.

Video – Subnet with the Magic Number

- This video will demonstrate subnetting with the magic number.

Refer IP Addressing eBook

- Subnetting at Page 20 – 27
- **Classless Interdomain Routing (CIDR)**
- <https://www.youtube.com/watch?v=Q1U9wVXRuHA>
- **Subnetting in table method**
- <https://www.youtube.com/watch?v=ecCuyq-Wprc>

Packet Tracer – Subnet an IPv4 Network

In this Packet Tracer, you will do the following:

- Design an IPv4 Network Subnetting Scheme
- Configure the Devices
- Test and Troubleshoot the Network